

Trainings ▾

General Information

CAUTION

The fan belt is mounted with spring tension. To reduce the possibility of personal injury from being pinched between the fan hub and fan idler, always release the spring tension before removing or installing the belt.

CAUTION

Always replace the complete bearing kit when any piece of the idler pulley bearing assembly requires replacement.

CAUTION

Never grease the bearings excessively. Too much grease will cause excessive heating due to the agitation of the grease. Excessive agitation of the grease will result in failure.

The fan belt is tensioned by the spring loaded idler assembly and does **not** require adjustment.

The kit includes a retaining ring, an inner spacer, two bearings, and two bearing races.

When the pivot arm cap is aligned properly with the spring, tension is a minimum of 68 N•m [50 ft-lb].

A grease nipple is standard on the pivot arm. The pivot arm **must** be greased at each scheduled maintenance interval (recommended maintenance interval refers to every 250 hours, or 6 months of operation).

Newer engines have a turnbuckle assembly to limit the travel of the idler pulley. Older engines have a shock absorber to limit the travel of the idler pulley. The shock absorber can be replaced by the turnbuckle assembly. The adjustment of the turnbuckle **must** be checked at each scheduled maintenance interval (recommended maintenance interval refers to every 250 hours, or 6 months of operation). The shock absorber **must** be checked for fluid leakage and loss of vibration absorption at each scheduled maintenance interval (recommended maintenance interval refers to every 250 hours, or 6 months of operation).

Fan hubs and supports are available in various drive ratios and fan center locations. When replacing the fan hub, **always** check the part number. Make sure the replacement is compatible.

All of the belt driven fan hubs contain anti-friction bearings that are the tapered roller type. Bearing end clearance is controlled by the use of inner and outer bearing spacers. It is a good service practice to tag the bearings for location during removal. **Always** replace the bearing and the bearing race if either piece requires replacement.

The idler pulley contains two anti-friction bearings that are the tapered roller type. The bearing end clearance is properly set by matching select components in the kit.

The following is the proper method of applying water pump type grease to the fan hub or idler pulley bearings. Do **not** use lithium base grease for fan hub bearings.

Pack both bearings with grease. Fill the cavity between the bearings 2/3-full of grease. Fill the cavity between the front bearing and the end of the hub 2/3-full of grease.

Preparatory Steps

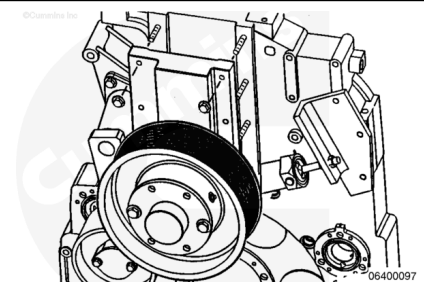
- Remove the fan. Refer to Procedure 008-040 in Section 8. (</qs3/pubsys2/xml/en/procedures/20/20-008-040.html>)
- Remove the fan belt. Refer to Procedure 008-002 in Section 8. (</qs3/pubsys2/xml/en/procedures/20/20-008-002-tr.html>)



Remove

⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.

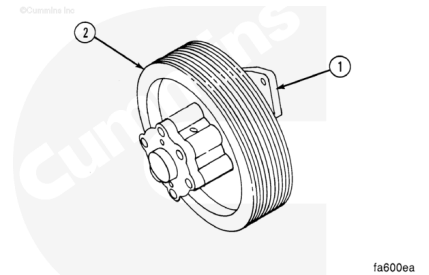


Remove the eight nuts and washers, and the fan hub.

Clean and Inspect for Reuse

⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.



⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

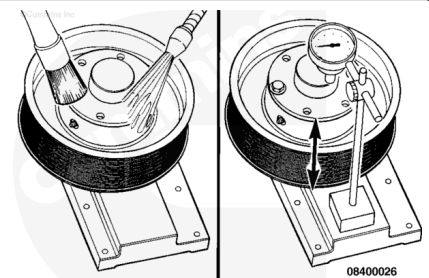
Use solvent to clean the fan hub.

Dry with compressed air

Inspect the fan hub for reuse. Rotate the pulley to check for rough or damaged bearings.

Measure bearing end clearance.

Bearing End Clearance		
mm		in
0.03	MIN	0.001
0.25	MAX	0.010



Inspect the grease seals for damage.

If the clearance is **not** within the specifications, the fan hub **must** be rebuilt.

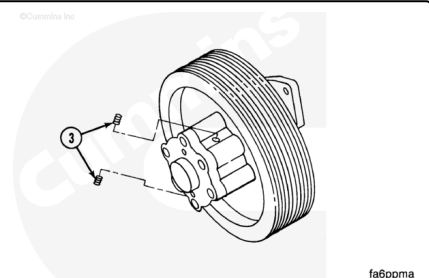
Rotate the pulley to check for rough or damaged bearings.

If the fan hub does **not** rotate freely, the fan hub **must** be rebuilt.

Remove the pipe plugs (3). Install a grease fitting in one of the holes.

Use a grease gun and water pump type grease. Pump grease into the hub until it begins to come out of the open plug hole.

Remove the grease fitting. Install the pipe plugs (3).

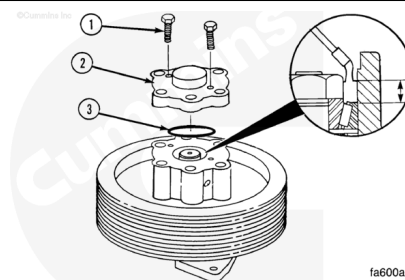


Torque Value: 15 n•m [135 in-lb]

Disassemble

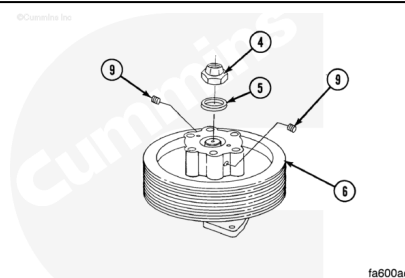
Remove the capscrews from the fan pilot.

Remove the fan pilot and discard o-ring.

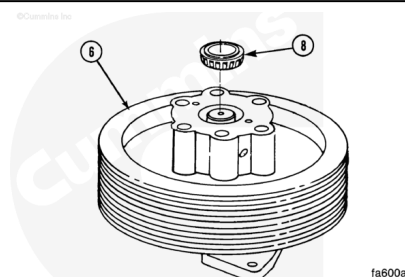


Remove the locknut (4).

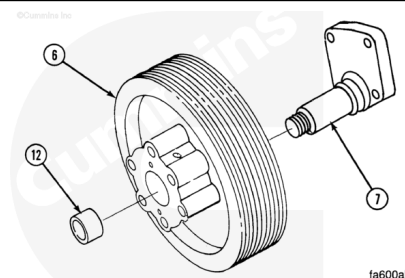
Remove the hardened washer (5).



Remove the outer fan hub bearing (8).

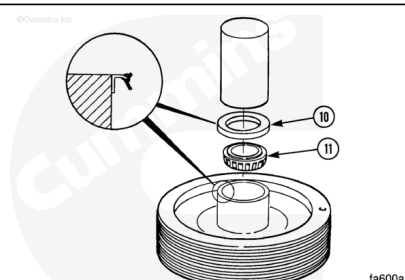


Remove the inner bearing spacer (12) from the fan hub (6) and shaft (7).



Remove the fan hub seal (10) and the inner bearing (11).

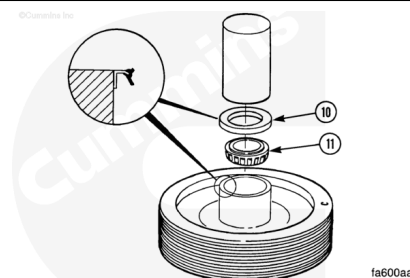
Use a pry bar to remove the seal. Do **not** damage the fan hub bore



Assemble

⚠ CAUTION ⚠

Do not lubricate the seal surface of the shaft. The seal and seal surface must be clean and dry.



Note : Excessive grease will affect the end float reading.

Lightly grease the inner bearing (11) with Mobiplex® 48 grease, or equivalent.

Install the bearing in the hub

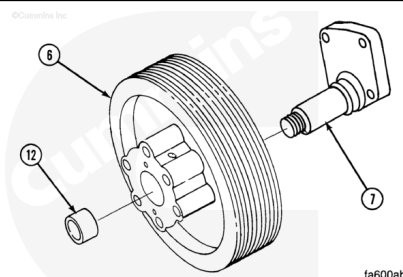
Apply Loctite® 675, or equivalent, to the outside diameter of the seal.

Use mandrel to install the seal (10) with the flat unseamed face toward the shoulder of the driver, press the seal in until it bottoms.

Carefully install the fan hub seal end first over the shaft, ensuring no damage occurs to the seal. The rear bearing **must** slide easily on the shaft. If it does **not**, check for chips or burrs.

Install the inner bearing spacer (12). The following spacers are available to provide correct end clearance when the fan hub has been assembled.

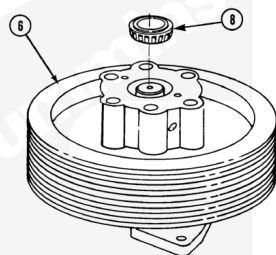
Reference the following table for fan hub styles:



Two Spacer Fan Hubs	Other Fan Hubs
3627681	3627677
3627682	3627678
3627683	3627679
3627684	3627680
N/A	3628830

Lightly grease the front bearing (8).

Install the bearing, it **must** slide easily over the shaft. If it does **not**, check for chips and burrs.



fa600ac

⚠ CAUTION ⚠

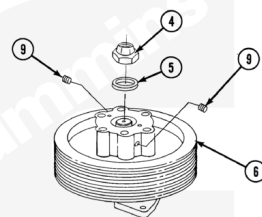
Remove all grease from the threads on the shaft before installing the lock nut. Grease on the threads reduce the torque retention of the lock nut.

Install the hardened washer (5).

Install the lock nut (4).

Tighten the lock nut.

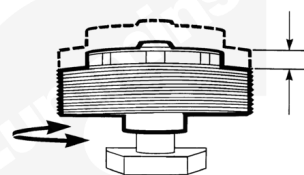
Torque Value: 205 n•m [150 ft-lb]



fa600ad

Use a dial indicator to measure the bearing end clearance.

Bearing End Clearance		
mm		in
0.025	MIN	0.001
0.102	MAX	0.004



fa600la

If the clearance is **not** within specifications, rebuild the fan hub with a suitable size spacer. Measure the bearing end clearance.

⚠ CAUTION ⚠

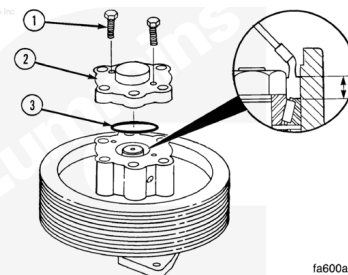
Do not use more grease than specified. Too much grease will cause excessive heating and damage.

Use Mobiplex® 48 grease, or equivalent, and fill the cavity with 120 cc [5 oz] of grease.

Install the o-ring (3) on the fan pilot (2).

Use vegetable oil to lubricate the o-ring.

Install the fan pilot and o-ring.



fa600ae

Install the capscrews (1).

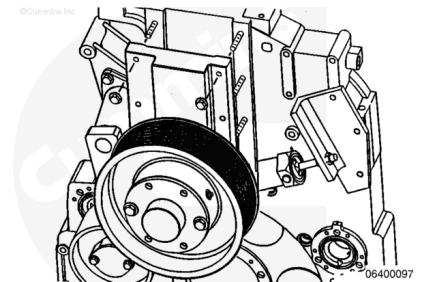
Tighten the capscrews.

Torque Value: 20 n•m [177 in-lb]

Install

⚠ WARNING ⚠

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Install the fan hub, the eight washers, and nuts.

Tighten the fan hub nuts.

Torque Value: 34 n•m [25 ft-lb]

Finishing Steps

- Install the fan belt. Refer to Procedure 008-002 in Section 8. (</qs3/pubsys2/xml/en/procedures/20/20-008-002-tr.html>)
- Install the fan. Refer to Procedure 008-040 in Section 8. (</qs3/pubsys2/xml/en/procedures/20/20-008-040.html>)
- Adjust the belt tensioner. Refer to Procedure 008-029 in Section 8. (</qs3/pubsys2/xml/en/procedures/20/20-008-029-tr.html>)
- Operate the engine and check for noise.

